

Introduction to Data Science:

Definition:

Data science is all about using data to solve problems. The problem could be decision making such as identifying which email is spam and which is not.

Data science is the practice of mining large data sets of raw data, both structured and unstructured, to identify patterns and extract actionable insight from them.

Data science is the field of study that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data. Data science practitioners apply machine learning algorithms to numbers, text, images, video, audio, and more to produce artificial intelligence (AI) systems to perform tasks that ordinarily require human intelligence. In turn, these systems generate insights which analysts and business users can translate into tangible business value.

Big data and data science hype:

Big Data:

Big data is a field that treats ways to analyse, systematically extract information from, or otherwise deal with data sets that are too large or complex to be dealt with by traditional data-processing application software.

Or

Big data is a blanket term for any collection of data sets so large or complex that it becomes difficult to process them using traditional data management techniques.

data science hype

With the development of science and technology, the scale of data available to human society is growing fast. From day to day, large amounts of data are generated and stored at all times.

The frequent overuse of "AI" when referring to any solution that makes any kind of prediction has been a major cause of this hype. Because of frequent overuse, people instinctively associate data science projects with near perfect human-like autonomous solutions. Or, at a minimum, people perceive that data science can easily solve their specific predictive need, without any regard to whether their organizational data will support such a model.

Why Data Science:

- Data science is used almost everywhere in both commercial and non-commercial settings.
 - Commercial companies in almost every industry use data science
 - to gain insights into their customers, processes, staff, and products
 - use data science to offer customers a better user experience
 - cross-sell, up-sell, and personalize their offerings

e.g.: - Google AdSense, Moneyball-people analytics.
- Financial institutions use data science
 - to predict stock markets,
 - determine the risk of lending money, and
 - learn how to attract new clients for their services.
- Ex: At least 50% of trades world wide are performed automatically by machines based on algorithms developed by data scientists with the help of big data and data science techniques.
- A data scientist in a governmental organization gets to work on diverse projects such as detecting fraud and other criminal activity or optimizing project funding. Ex: data.gov
- Non-governmental agencies use it to raise money and defend their causes.

- The World Wildlife Fund (WWF), for instance, employs data scientists to increase the effectiveness of their fundraising efforts.
- Universities use data science in their research but also to enhance the study experience of their students.
- The rise of massive open online courses (MOOC) produces a lot of data, which allows universities to study how this type of learning can complement traditional classes.

Getting Past the Hype:

Data science is reaching the end of its hype cycle and the skills for becoming a data scientist are changing. Besides of fiddling about the best performant machine learning model, it is more important than ever to make data science with business impact. the current skill set that help define and build up a data science workflow delivering real value.

Data Science applications in the business world work in a similar way. Data Scientists have to design models, which might be adaptive, sophisticated and able to learn over time. If you are working with experienced Data Scientists they will be able to encode knowledge about how your business works into their models, which in turn might lead to more accurate predictions. At the end of the day any time you are tackling a very specialized task which requires specialized industry knowledge, the adaptation is not automatic, instead the underlying predictive models have to be custom-tailored by a Data Scientist to include this domain knowledge.

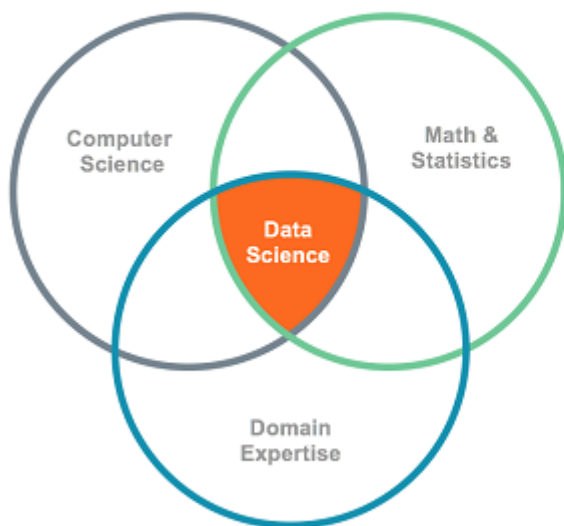
The current landscape:

Data Scientists are currently the driving force in many companies' AI strategy, and the demand for them has significantly risen in recent years.

Data Science						
Discipline	Analytics		Statistics		Machine Learning	
Focus	"Ask better questions"		"Give better answers"		"Decision making at scale"	
Sub-Discipline	Business Analysis	Exploratory Data Analytics	Descriptive Statistics	Inferential Statistics	Supervised Machine Learning	Unsupervised Machine Learning
Tools	Spreadsheets, Visualization Tools		Statistical Packages		Programming Languages	
Process	Business Understanding > Data Collection > Data Preparation > Model Building > Evaluation > Deployment					

Or

Data science is a powerful combination of various disciplines.



Computer Science Skills

- Programming
- Big data technologies

Math and Statistics Knowledge

- Machine learning
- Ensemble models
- Anomaly detection

Domain Expertise

- Business knowledge
- Expert systems
- User testing

A key aspect of data science is the utilization of the scientific method to form and challenge hypotheses to validate conclusions about underlying patterns in data.

Practicing data science requires the combining of a diverse set of skills. Data scientists need to be able to query and manipulate large swaths of data, so a strong computer science background is a must. Additionally, familiarity with mathematics and statistics help form a strong understanding of the algorithms commonly deployed and tuned.

Who is data scientist:

- A data scientist is a professional responsible for collecting, analyzing and interpreting extremely large amounts of data.
- The data scientist role is an offshoot of several traditional technical roles, including mathematician, scientist, statistician and computer professional. This job requires the use of advanced analytics technologies, including machine learning and predictive modelling.
- A data scientist requires large amounts of data to develop hypotheses, make inferences, and analyse customer and market trends. Basic responsibilities include gathering and analysing data, using various types of analytics and reporting tools to detect patterns, trends and relationships in data sets.
- Data scientists are analytical experts who utilize their skills in both technology and social science to find trends and manage data. They use industry knowledge, contextual understanding, skepticism of existing assumptions – to uncover solutions to business challenges.
- A data scientist's work typically involves making sense of messy, unstructured data, from sources such as smart devices, social media feeds, and emails that don't neatly fit into a database.